testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Progression

Progression

Arithmetic Progression (AP)

- **General Term :** $a_n = a + (n - 1)d$
- **Sum of the First n Terms of an AP**

  $S_n = \frac{n}{2}[2a + (n - 1)d]$ Or $S_n = \frac{n}{2}(a + l)$

- **Arithmetic mean** = $\frac{x_1+x_2+x_3+,...,+x_n}{n}$

  ➔ For two numbers a, b, the arithmetic mean (AM) is given by

  AM = $\frac{a+b}{2}$

- **Inserting 'n' AMs between 'a' and 'b'**

  ➔ Last term , b = a + (n + 1)d

  ➔ Common difference, d = (b-a)/(n+1)

Important results of AP

- Common difference can be zero, + ve or – ve.

  ➔ If d > 0 ⇒ increasing AP and the sequence tends to +∞

  ➔ If d < 0 ⇒ decreasing AP and the sequence tends to –∞

  ➔ If d = 0 ⇒ constant AP (all the terms remain same)

- The nth term from end is (m – n + 1) term from the beginning, where m is the total number of terms and is given by the following expression:

  Tm-n+1 = Tm – (n–1) d

- If the mth term is n and the nth term is m, then the (m + n)th term is 0.
- If m times the mth term is equal to n times the nth term, then the (m + n)th term is 0.

Geometric Progression (GP)

- **General Term :** $a_n = ar^{n-1}$

- **Sum of the First n Terms of an AP**

$S_n = a\frac{r^n - 1}{r - 1}$, [for r > 1]

Or

$S_n = a\frac{r^n - 1}{r - 1}$, [for r < 1]

Or

$S_n = n \cdot a$, [for $r = 1$]

- *For an infinite GP where* $|r| < 1$, *the sum is* :

$$S_\infty = \frac{a}{1 - r}$$

- **Geometric mean =** $\sqrt[n]{a_1 * a_2 * a_3 * \ldots * a_n}$

➔ For two numbers a and b, the geometric mean is given by :

GM = $\sqrt{ab}$

- **Inserting 'n' GMs between 'a' and 'b'**

➔ Last term , b = $ar^{(n+1)}$

$\Rightarrow r = \left(\frac{b}{a}\right)^{\frac{1}{(n+1)}}$

Relation between A.M. and GM

For any two non-negative numbers, A.M. ≥ G.M.

Note : Equality for AM, G.M. (i.e. A.M. = GM) exists when a = b.

Arithmetic-Geometric Progression (AGP)

- **General Term :** $A_n = [a + (n - 1)d]\, br^{(n-1)}$

Harmonic Progression (HP)

- **General Term :** However, the terms of an HP are the reciprocals of the terms of an Arithmetic Progression.

  $a,\ (a + d),\ (a + 2d),\ (a + 3d)\ldots$ are the terms of an AP,

  $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \frac{1}{a+3d} \ldots$ are the terms of the corresponding HP.

  Note : 0 cannot be a term of H.P. because ∞ is not a term of AP, but ∞ can be a term of HP.

- **Sum of the First n Terms of an HP:**

$$S_n = \frac{1}{d} ln \left\{ \frac{2a+(2n-1)d}{2a-d} \right\}$$

- **Geometric mean:** HM = n / $[(1/x_1) + (1/x_2) + (1/x_3) + \ldots + (1/x_n)]$

  The Harmonic mean of two terms a and b = (2ab) / (a + b).

  The Harmonic mean of three terms a, b, and c = (3abc)/(ab + bc + ca)

Relationship Between AM, GM, HM

- AM > GM > HM
- **GM^2 = AM x HM ,**

  where A.M, G.M, and H.M are in G.P